

Customer Shopping Behavior Analysis

Project Overview

This project analyzes customer shopping behavior using **Python, MySQL, and Power BI**. The workflow begins with data preprocessing and exploratory data analysis (EDA), followed by feature engineering and data cleaning. The processed dataset is then imported into MySQL to perform business-focused SQL queries. Finally, the data is connected to Power BI to build an interactive dashboard that presents KPIs, customer trends, sales performance, and purchasing insights. This report demonstrates the complete end-to-end data analytics workflow from raw data preparation to business intelligence reporting.

Tools & Technologies

Python (Pandas, NumPy) • MySQL • Power BI

Customer Shopping Behavior Analysis

Python Data Preprocessing & Exploratory Data Analysis

Step 1: Importing Libraries & Loading Dataset

Imported pandas and numpy, loaded the CSV dataset using `pd.read_csv()`, and displayed the first 10 rows using `df.head(10)`.

```
[6]: import pandas as pd
import numpy as np

[7]: df = pd.read_csv("../customer_shopping_behavior.csv")

[8]: df.head(10)
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	P I
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	
5	6	46	Male	Sneakers	Footwear	20	Wyoming	M	White	Summer	2.9	Yes	Standard	Yes	Yes	14	
6	7	63	Male	Shirt	Clothing	85	Montana	M	Gray	Fall	3.2	Yes	Free Shipping	Yes	Yes	49	
7	8	27	Male	Shorts	Clothing	34	Louisiana	L	Charcoal	Winter	3.2	Yes	Free Shipping	Yes	Yes	19	
8	9	26	Male	Coat	Outerwear	97	West Virginia	L	Silver	Summer	2.6	Yes	Express	Yes	Yes	8	
9	10	57	Male	Handbag	Accessories	31	Missouri	M	Pink	Spring	4.8	Yes	2-Day Shipping	Yes	Yes	4	

Step 2: Descriptive Statistics

Used `df.describe()` to summarize numerical columns including count, mean, standard deviation, quartiles, minimum and maximum values.

```
df.describe()
```

	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
count	3900.000000	3900.000000	3900.000000	3863.000000	3900.000000
mean	1950.500000	44.068462	59.764359	3.750065	25.351538
std	1125.977353	15.207589	23.685392	0.716983	14.447125
min	1.000000	18.000000	20.000000	2.500000	1.000000
25%	975.750000	31.000000	39.000000	3.100000	13.000000
50%	1950.500000	44.000000	60.000000	3.800000	25.000000
75%	2925.250000	57.000000	81.000000	4.400000	38.000000
max	3900.000000	70.000000	100.000000	5.000000	50.000000

Step 3: Missing Value Check

Checked missing values using `df.isnull().sum()`. Only the Review Rating column contained 37 missing values.

```
df.isnull().sum()

Customer ID      0
Age              0
Gender           0
Item Purchased   0
Category         0
Purchase Amount (USD)  0
Location         0
Size            0
Color           0
Season          0
Review Rating    37
Subscription Status  0
Shipping Type    0
Discount Applied  0
Promo Code Used  0
Previous Purchases  0
Payment Method   0
Frequency of Purchases  0
dtype: int64
```

Step 4: Column Standardization

Converted column names to lowercase, replaced spaces with underscores and renamed `purchase_amount_(usd)` to `purchase_amount`.

```
: df.columns=df.columns.str.lower()
df.columns =df.columns.str.replace(" ", "_")

: df.columns

: Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
      'purchase_amount_(usd)', 'location', 'size', 'color', 'season',
      'review_rating', 'subscription_status', 'shipping_type',
      'discount_applied', 'promo_code_used', 'previous_purchases',
      'payment_method', 'frequency_of_purchases'],
      dtype='object')

: df=df.rename(columns={'purchase_amount_(usd)': 'purchase_amount'})

: df.columns

: Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
      'purchase_amount', 'location', 'size', 'color', 'season',
      'review_rating', 'subscription_status', 'shipping_type',
      'discount_applied', 'promo_code_used', 'previous_purchases',
      'payment_method', 'frequency_of_purchases'],
      dtype='object')
```

Step 5: Creating Age Groups

Created the age_group feature using pd.qcut() to divide customers into four age categories.

```
# categorizes Ages group
labels=["Young Adult","Adult","Middle-aged","Senior"]
df["age_group"]=pd.qcut(df["age"], q=4, labels =labels)
```

```
df[["age","age_group"]].head(5)
```

	age	age_group
0	55	Middle-aged
1	19	Young Adult
2	50	Middle-aged
3	21	Young Adult
4	45	Middle-aged

Step 6: Purchase Frequency Mapping

Mapped purchase frequency text values into numeric day values and created purchase_frequency_days.

```
# create column purchase_frequency_days

frequency_mapping = {
    'Fortnightly': 14,
    'Weekly': 7,
    'Monthly': 30,
    'Quarterly': 90,
    'Bi-Weekly': 14,
    'Annually': 365,
    'Every 3 Months': 90
}

df['purchase_frequency_days'] = df['frequency_of_purchases'].map(frequency_mapping)
```

Step 7: Removing Redundant Column

Verified discount_applied and promo_code_used contained identical values, then removed the redundant column.

```
(df["discount_applied"] == df["promo_code_used"]).all()
```

```
np.True_
```

```
df= df.drop("promo_code_used",axis=1)
```

```
df.columns
```

```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
      'purchase_amount', 'location', 'size', 'color', 'season',
      'review_rating', 'subscription_status', 'shipping_type',
      'discount_applied', 'previous_purchases', 'payment_method',
      'frequency_of_purchases', 'age_group', 'purchase_frequency_days'],
      dtype='object')
```

Additional Python Data Preparation Summary

- **Handling Missing Values**

Checked the dataset for missing records and filled the missing values in the **Review Rating** column using the median value for each product category.

- **Column Standardization**

Converted all column names to lowercase, replaced spaces with underscores, and simplified lengthy names to improve readability and maintain consistency.

- **Feature Engineering**

Created the **age_group** column to classify customers into age categories and the **purchase_frequency_days** column to convert purchase frequency labels into numerical day values.

- **Data Consistency Check**

Compared the **discount_applied** and **promo_code_used** columns. Since both contained the same information, the redundant **promo_code_used** column was removed.

- **MySQL Database Integration**

After completing all preprocessing and transformation tasks, the cleaned DataFrame was successfully connected to the **MySQL** database and loaded for SQL queries and business analysis.

Customer Shopping Behavior Analysis

MySQL Query Results & Power BI Dashboard

1. Revenue by Gender

Calculated the total purchase revenue generated by male and female customers. The result shows that male customers contributed higher overall revenue.

gender	revenue
Male	157890
Female	75191

2. High-Spending Discount Customers

Retrieved customers who used discounts while spending more than the average purchase amount.

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72

customer 3 x

3. Top 5 Products by Average Rating

Found the five products with the highest average customer review ratings.

item_purchased	avg_rating
Gloves	3.8614285714285725
Sandals	3.8443750000000003
Boots	3.8187500000000005
Hat	3.8012987012987005
Skirt	3.784810126582278

4. Average Purchase by Shipping Type

Compared the average purchase amount across different shipping methods.

shipping_type	avg_pur_amount
Express	60.4752
Standard	58.4602

5. Subscribers vs Non-Subscribers

Compared customer count, average spending, and total revenue based on subscription status.

avg_spent	total_rev	subscription_status	total_customer
59.4919	62645	Yes	1053
59.8651	170436	No	2847

6. Products with Highest Discount Rate

Identified products receiving the highest percentage of discounted purchases.

item_purchased	total_purchases	discount_rate
Hat	154	50.00
Sneakers	145	49.66
Coat	161	49.07
Sweater	164	48.17
Pants	171	47.37

7. Customer Segmentation

Grouped customers into Loyal, Returning, and New categories based on purchase history.

customer_type	total_customers
Loyal	3116
Returning	701
New	83

8. Most Purchased Products

Displayed the products with the highest total purchase count.

most_purchase	total_purchase
Blouse	171
Pants	171
Jewelry	171

9. Repeat Buyers by Subscription

Counted repeat buyers according to their subscription status.

subscription_status	total_customers
Yes	958
No	2518

10. Revenue by Age Group

Calculated the total revenue generated by each customer age group.

age_group	total_revenue
Young Adult	62143
Middle-aged	59197
Adult	55978
Senior	55763

Power BI Dashboard

Created an interactive dashboard to visualize customer behavior. The dashboard displays KPI cards for total customers, average review rating, and average purchase amount, along with visuals showing subscription distribution, revenue by category, sales by category, sales by age group, and revenue by age group. Slicers allow users to filter the report by subscription status, gender, category, and shipping type for interactive analysis.

Customer Behavior Dashboard

Subscription Status

- ☐ No
☐ Yes

Gender

- ☐ Female
☐ Male

Category

- ☐ Accessories
☐ Clothing
☐ Footwear
☐ Outerwear

shipping_type

- ☐ 2-Day Shipping
☐ Express
☐ Free Shipping
☐ Next Day Air
☐ Standard
☐ Store Pickup

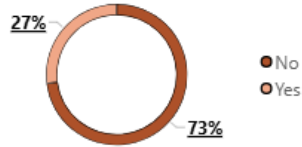
Customers
3.9K

Average Review
3.75

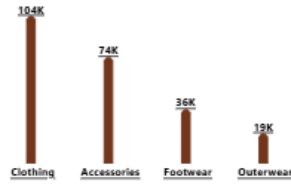
Average Purchase
\$59.76



Percentage of Subscription Status



Revenue by category



Sales by category



Sales By Age Group



Revenue by Age Group



Power BI Dashboard Explanation

The cleaned dataset was imported into **Power BI** to create an interactive dashboard for analyzing customer shopping behavior. KPI cards, charts, and slicers were used to present important business insights such as customer count, average purchase amount, review rating, sales, and revenue. The dashboard enables users to filter data by subscription status, gender, category, and shipping type for better decision-making.